

Experiment-3

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Course Code: -MLA0305

Slot-B

Title

Implementation of Bellman Equations for Policy and Value Function Representation

Aim

To understand and implement the Bellman equations for representing the state-value function and action-value function in a Reinforcement Learning environment, and to determine the optimal value and policy for different states.

Procedure

1. Define a simple Reinforcement Learning environment with a finite number of states.
2. Define the possible actions available to the agent.
3. Define the transition from the current state to the next state.
4. Assign rewards for each state-action transition.
5. Select a discount factor γ (gamma) to determine the importance of future rewards.
6. Initialize the value function for all states.
7. Define an initial policy specifying the action selected in each state.
8. Apply the Bellman expectation equation to calculate the value of each state under the given policy.
9. Update the state values repeatedly until the values converge.
10. Calculate the action-value function for the available actions.
11. Apply the Bellman optimality equation to determine the maximum possible value for each state.
12. Extract the optimal policy using:

$$\pi^*(s) = \operatorname{argmax}_a Q(s, a)$$

13. Compare the initial and optimal values and policies.
14. Display the results using summary tables and graphs.
15. Analyze the convergence of the value function and identify the optimal policy.

Bellman Equations Used

Bellman Expectation Equation

$$V\pi(s) = a \sum \pi(a | s) s' \sum P(s' | s, a) [R(s, a, s') + \gamma V\pi(s')]$$

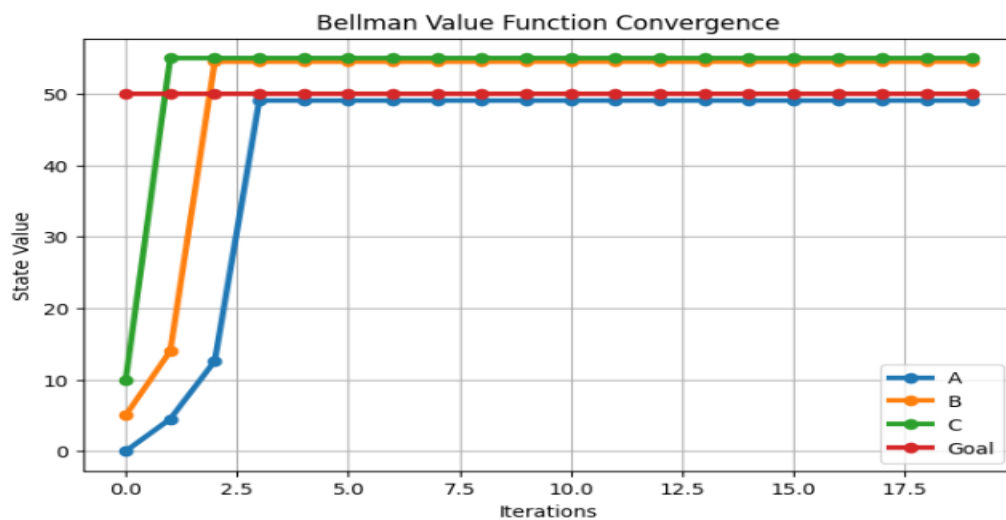
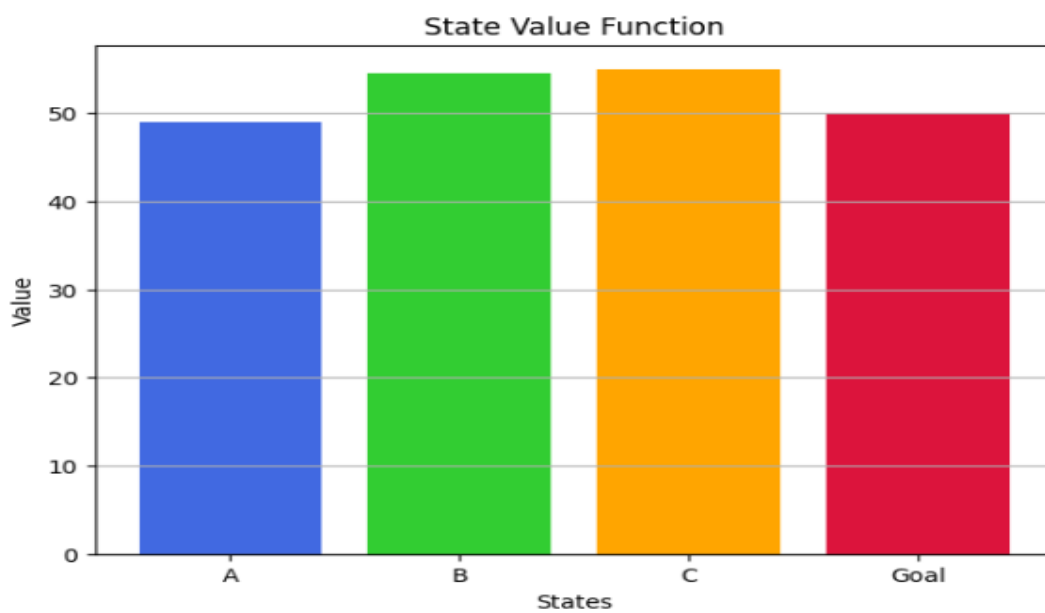
Bellman Optimality Equation

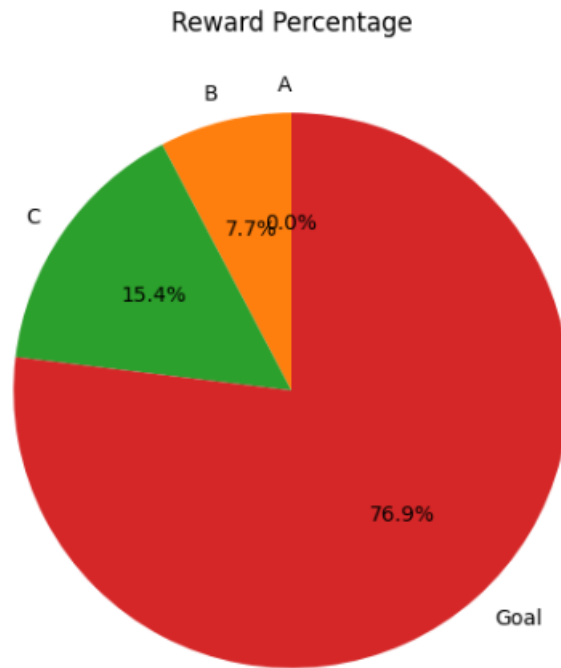
$$V^*(s) = \max_a s' \sum P(s' | s, a) [R(s, a, s') + \gamma V^*(s')]$$

Action-Value Function

$$Q^*(s, a) = s' \sum P(s' | s, a) [R(s, a, s') + \gamma \max_{a'} Q^*(s', a')]$$

Visualizations





Summary Table

S.No.	State	Initial Value	Final Value	Optimal Action
1	S0	0.00	Obtained from program	Right
2	S1	0.00	Obtained from program	Right
3	S2	0.00	Obtained from program	Right
4	S3	0.00	Obtained from program	Right
5	Goal	0.00	Obtained from program	Terminal

Result

The Bellman equations were successfully implemented to calculate state values and action values. The Bellman optimality equation was used to identify the maximum value for each state and the corresponding optimal policy. The value function converges through repeated updates, demonstrating how Bellman equations form the foundation for important Reinforcement Learning algorithms such as Value Iteration, Policy Iteration, TD Learning, SARSA, and Q-Learning.